SSC CGL Arithmetic Formula

Average

Average $= \frac{Sum\ of\ magnitude\ of\ all\ Quantities}{Number\ of\ Quantities}$

Average of n numbers = $\frac{X1+X2+X3+X4+X5,.....Xn}{n}$

Important Properties (महत्वपूर्ण गुण)

- **In a set of numbers, average** always lies above the **lowest number** of the set and below the **highest number** of the set.
- If all the number of a given set are equal, then the average will also be the same as the equal number.
- If zero is one of the numbers in a given set, then zero should also be included while calculating the average.
- If each quantity of the set is increased by a certain value 'a' then average of the set will also be increased by 'a'

Some Important Results:

1. For Consecutive Natural Numbers

- Average of first n natural numbers: $\frac{n+1}{2}$
- Average of squares of first n natural numbers: $\frac{(n+1)(2n+1)}{6}$
- Average of cubes of first n natural numbers: $\frac{[n(n+1)^2]}{4}$

2. For Consecutive Even Numbers

- Average of first n even numbers: n+1
- Average of squares of first n even numbers: $\frac{2(n+1)(2n+1)}{3}$
- Average of cubes of first n even numbers: $2n(n+1)^2$

3. For Consecutive Odd Numbers:

- Average of first n odd numbers: n
- Average of squares of first n odd numbers: $\frac{(2n+1)(2n-1)}{3}$
- Average of cubes of first n odd numbers: $n(2n^2-1)$

Average Speed:

- If a person travels from P to Q at a speed of x km/h and returns from Q to P at a speed of y km/h, the average speed for the entire journey is:

Average speed = $\frac{2xy}{x+y}$

Or

Average speed = Total distance / Total time taken

- If a distance is traveled with three different speeds: a km/h, b km/h, and c km/h, the average speed for the total journey is:

Average speed = $\frac{3abc}{ab+bc+ca}$

When Data is Misread:

- If the average of n numbers is m, but later it was found that a number 'a' was misread as 'b', the correct average will be:

$Correct\ Average = m + \frac{a-b}{n}$

- If the average of n numbers is m, but later it was found that two numbers 'a' and 'b' were misread as 'p' and 'q', the correct average will be:

  $Correct\ Average\ =\ m\ +\frac{[(a\ +\ b)-\ (p\ +\ q)]}{n}$

Concept of Age:

- If in the group of N persons, a new person replaces a person of age T years, so that the average age increases by t years:

The age of the new person is: T + N * t

- If the average age decreases by t years after the entry of a new person:

The age of the new person is: T - N * t

- If the average age of a group of N students is T years, and n students join, causing the average age of the group to increase by t years, then:

The average age of the new students is: $T\ +\ [\left(\frac{N}{n}\right)\ +\ 1]\ *t$

Important Concept

- If in any series having common difference "d" and Average "k", "x" numbers are added in forward or backward, then

  New Avg. = k ± $(\frac{xd}{2})$
- In series of even or odd having Avg. "k", when we add "x" no. in forward or backward, Then

  New Avg. = k ± x
- In series of natural no. having Avg. "k", when we add "x" no. in forward or backward, Then

  New Avg. = k ± $(\frac{x}{2})$
- If the average of n_1 observations is a_1, the average of n_2 observations is a_2 and so on, then:

  Average of all the observations =$\frac{n^1a^1+n^2a^2+\ldots}{n^1+n^2+\ldots}$.
- If the average of n observations is a but the average becomes b when one observation is eliminated, then value of eliminated observation = n(a − b) + b.
- If the average of n observations is a but the average becomes b when a new observation is added, then value of added observation = n(b − a) + b.
- If the average of n students in a class is a, where average of passed students is x and average of failed students is y, then:

  Number of students passed = $\frac{n(a-y)}{x-y}$.
- Bowling Average = $\frac{Total\ runs\ given}{Total\ wickets\ taken}$
- Batting Average = $\frac{Total\ runs\ scored}{Total\ number\ of\ innings\ played}$
- Total runs = Bowling average × Wickets

  Or

Total average = $\frac{Bowling\ Average\ (1)\times Wicket\ (1)+\ Bowling\ Average\ (2)\times Wicket\ (2)}{Wicket\ (1)+\ Wicket\ (2)}$

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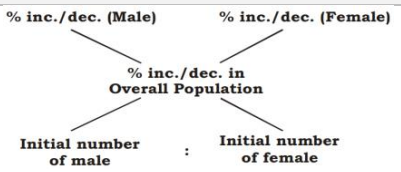
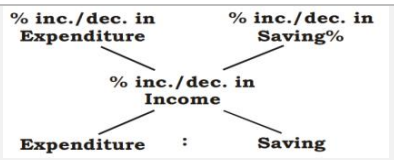
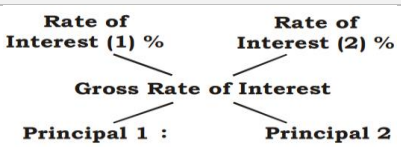
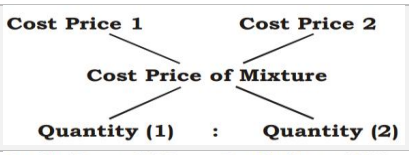
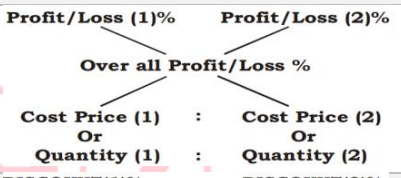
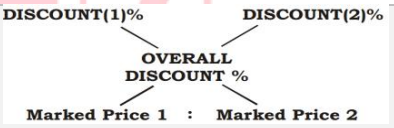
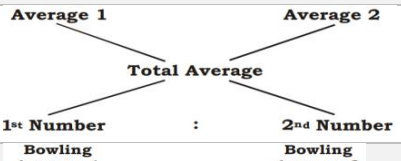
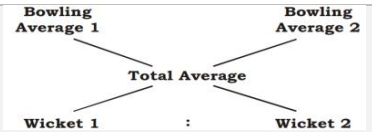
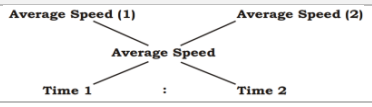
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Mixture and Allegation

Topic	Concept	Shortcut / Use of Allegation
Mixture	Mixing two quantities with known ratios to get a final ratio or price	
Percentage (Population)	% change in male & female population given, along with total % change	
Percentage (Income)	% change in income, expenditure, savings given	
Simple Interest	Two schemes with different interest rates and one total amount after interest	
Profit & Loss (CP based)	Cost prices of two items & SP of mixture + profit/loss % given	
Profit & Loss (Profit %)	Profit % from two items & overall profit % given	
Discount	Two discount %s on parts of a marked price & net discount % given	
Average	Average of two groups & overall average given	
Bowler's Average	Bowling average before and in one innings, and final average given	
Time & Distance	Two speeds for equal or unequal distances, and average speed given	

Number System

Unit Digit:

Concept	Explanation	Example
Unit Digit	The digit in the one's place of a number.	Unit digit of **237** is **7**.
Unit Digit of Powers	Follows a cyclic pattern depending on the base.	Unit digit of 21=22^1 = 221=2, 22=42^2 = 422=4, 23=82^3 = 823=8...
Cyclic Patterns	Most digits (1–9) have repeating cycles in their powers.	2 → {2,4,8,6} (Cycle length = 4)
Shortcut Method	Divide power by cycle length & find remainder → use remainder to find unit digit.	Find unit digit of 7457^{45}745: Cycle of 7 = {7,9,3,1}, 45 mod 4 = 1 → Ans: **7**
Special Cases	Some numbers (like 5, 6) always end in the same digit for all powers.	5n5^n5n ends in **5**, 6n6^n6n ends in **6**.

Factors:

Concept	Explanation	Example
Factor/Divisor	A number that divides another number exactly (no remainder).	Factors of **12**: 1, 2, 3, 4, 6, 12
Total No. of Factors	Use prime factorization: Add 1 to each exponent and multiply.	12=22×3112 = 2^2 × 3^112=22×31 → (2+1)(1+1) = **6** factors
Sum of Factors	Use formula with prime factorization.	12=22×3112 = 2^2 × 3^112=22×31 → (1+2+4)(1+3) = 7×4 = **28**
Even/Odd Factors	Even → include only factors with at least one 2; Odd → exclude all 2s.	Odd factors of 36 = factors of 32=1,3,93^2 = 1, 3, 932=1,3,9
Perfect Square Check	A number is a perfect square if all exponents in its prime factorization are even.	36=22×3236 = 2^2 × 3^236=22×32 → Perfect square ✅
Co-prime Numbers	Two numbers having only 1 as their common factor.	(8, 15) are co-prime → GCD = 1
Highest Common Factor (HCF)	Greatest number that divides both numbers.	HCF of 12 and 18 = **6**
Least Common Multiple (LCM)	Smallest number divisible by both numbers.	LCM of 4 and 6 = **12**

No of Zeros:

Concept	Formula / Rule	Explanation / Example
Trailing zeros in $n!$	$\lfloor \frac{n}{5} \rfloor + \lfloor \frac{n}{25} \rfloor + \lfloor \frac{n}{125} \rfloor + \dots$	Count of 5s in prime factorization of $n!$; 10 = 2 × 5, but 2s are always more, so count 5s. **Eg:** Trailing zeros in $100!$ = 20 + 4 = **24**
Trailing zeros in a power a^b	Count of 10s in a^b	Prime factorize a, find min(power of 2, power of 5) in result
Trailing zeros in a number like 1000	Count number of zeros at the end	Eg: 1000 → 3 trailing zeros
Zeros in decimal	Depends on denominator of fraction	Eg: 1/10 = 0.1 (1 zero), 1/100 = 0.01 (2 zeros)
Ending zeros in product of numbers	Count net power of 10 in the product	Eg: $2^3 \times 5^4$ → Min(3,4) = 3 → **3 zeros**

Remainder Theorem:

Type of Division	Formula / Concept	Example
$f(a)$ is remainder when polynomial $f(x)$ is divided by $x-a$	Remainder = $f(a)$	Eg: Remainder when $f(x)=x^2+3x+2$ divided by $x-2$ is $f(2)=4+6+2=**12**$
$a^n \mod b$ when a and b are coprime	Use **Euler's Theorem:** $a^{\phi(b)} \equiv 1 \mod b$	Eg: $7^{100} \mod 10$: φ(10)=4 → $7^4 \equiv 1 \mod 10 \rightarrow 7^{100} \equiv 1^{25}=1$
$a^n \mod b$ for small values	Use cyclicity / patterns	Eg: $3^n \mod 4$ → 3, 1, 3, 1... pattern of 2
Divisibility of large numbers	Break number using modulo	Eg: Find remainder when 111...111 (15 times) is divided by 9: Since sum of digits = 15 → remainder = **6**
Division of powers like $2^{20} \mod 7$	Use cyclicity of powers	Powers of 2 mod 7: 2, 4, 1 (repeat every 3) → 20 mod 3 = 2 → answer is $2^2=**4**$
Special case: a^n-b^n divisible by $a-b$	Always divisible	Eg: 7^5-3^5 divisible by $7-3=4$

Divisibility Rules:

Divisor	Divisibility Rule	Example
2	If the number ends in 0, 2, 4, 6, or 8.	134 **is** divisible by 2 (ends in 4).
3	If the sum of the digits is divisible by 3.	123 → 1+2+3=6 → divisible by 3.
4	If the last **two digits** form a number divisible by 4.	316 **is** divisible by 4 (16 ÷ 4 = 4).
5	If the number ends in **0 or 5**.	475, 900 both end in 5 or 0.
6	If the number is divisible by **both 2 and 3**.	132 is even and 1+3+2=6 (div. by 3).
7	Double the last digit, subtract from the rest; if result is divisible by 7.	203: 20 – 6 = 14 → divisible by 7.
8	If the last **three digits** form a number divisible by 8.	1216 → 216 ÷ 8 = 27.
9	If the sum of the digits is divisible by 9.	729 → 7+2+9=18 → divisible by 9.
10	If the number ends in **0**.	370, 800 end in 0.
11	Alternating sum of digits is divisible by 11 (e.g. +, –, +, –...).	121 → 1–2+1 = 0 → divisible by 11.
12	If the number is divisible by both **3 and 4**.	240: 2+4+0=6 (div. by 3), 40 div. by 4.
13	Multiply last digit by 9, subtract from rest. If result divisible by 13.	637: 63 – (7×9)=63–63=0 → OK.
14	If the number is divisible by **2 and 7**.	196 is even and divisible by 7.
15	If the number is divisible by **3 and 5**.	225 is div. by 3 (2+2+5=9) and ends 5.
16	If the last **4 digits** form a number divisible by 16.	65472 → 5472 ÷ 16 = 342.
17	Subtract 5 × last digit from the rest. Repeat if needed.	221 → 22 – 5×1 = 17 → divisible.
18	If the number is divisible by **2 and 9**.	198 → even, 1+9+8=18 → divisible.
19	Multiply last digit by 2, add to rest. If divisible by 19.	133 → 13 + (2×3) = 13+6 = 19 → OK.
20	If the number ends in **00, 20, 40, 60, or 80**.	560 ends in 60 → divisible by 20.

Partnership

S. No.	Concept / Formula	Explanation	Example
1	**Types of Partnership** • Simple • Compound	- Simple: same time-Compound: different times	**Simple:** A & B invest ₹5000 each for 12 months → equal profit share.**Compound:** A invests for 12 months, B for 6 months → profits not equally divided.
2	**Profit Ratio** Profit ∝ Capital × Time	Multiply capital and time to find profit ratio	A: ₹6000 for 12 months → 6000×12 = 72000B: ₹8000 for 6 months → 8000×6 = 48000Ratio = 72000:48000 = **3:2**
3	**Individual Share of Profit** Partner's share = Total Profit × Ratio share	Apply the ratio on total profit	Total Profit = ₹2500Profit ratio A:B = 3:2A's share = ₹2500 × 3/5 = ₹1500B's share = ₹1000
4	**Capital when Profit & Time are known** Capital ∝ Profit / Time	Reverse use of the profit formula	A & B share profit in 4:5Time = 6 months (same)So, Capital ratio A:B = 4:5
5	**Equivalent Capital** Capital × Time = Equivalent Capital	Used to make fair comparisons	A: ₹2000 for 10 months = 20000B: ₹2500 for 8 months = 20000 → Equal profits
6	**Incoming Partner's Goodwill Share** Goodwill Share = Share × Total Goodwill	Used when a new partner pays to join	Goodwill = ₹60,000C gets 1/4th share⇒ C pays 1/4 × 60,000 = **₹15,000**
7	**Change in Capital** Recalculate capital after changes	Track changes in investment periods	A invests ₹4000 for 4 months, then ₹6000 for 8 months Total = (4000×4) + (6000×8) = **16,000 + 48,000 = ₹64,000**
8	**Sleeping Partner**	Doesn't participate in work, only capital	A (working), B (sleeping). Both invest equally for 12 months → Equal profit if no salary.
9	**Working Partner** May receive salary or commission	Active in management & gets fixed benefits	A & B are equal partnersA gets ₹200/month as salary for 12 months = ₹2400Subtract ₹2400 from profit, rest is shared.
10	**Salary to Working Partner** Deduct salary before profit division	Profit is adjusted before sharing	Profit = ₹12,000A gets salary ₹2000 Remaining ₹10,000 shared in 3:2 ratio

S. No.	Concept / Formula	Explanation	Example
11	**Average Capital** Use: Σ(Capital × Time)	Use when investments vary over time	A invests ₹5000 for 6 months + ₹7000 for 6 months Total = (5000×6) + (7000×6) = ₹72,000
12	**New Partner Admission** New Share = (Capital × Time) / Total	Calculate adjusted profit ratio after joining	A: ₹6000 for 12 months → 72000B: ₹4000 for 12 months → 48000C joins with ₹5000 for 6 months → 30000 New ratio A:B:C = 72000:48000:30000 = **12:8:5**

Partnership (Working and Sleeping Partners)

No.	Concept	Formula	Example (with Solution)
1	**Profit Sharing Ratio**	Partner's Share = (Capital × Time) / Total (Capital × Time)	A invests ₹20,000, B invests ₹30,000 for 12 months. Profit = ₹25,000 ➤ A:B = (20k×12):(30k×12) = 240k:360k = 2:3 ➤ A's share = ₹10,000; B's share = ₹15,000
2	**Salary to Working Partner**	1. Pay Salary/Commission First2. Divide Remaining Profit as per Capital × Time	A and B invest ₹25,000 each. A is working and gets ₹2,000/month for 4 months. Profit = ₹14,000 ➤ Salary to A = ₹8,000 ➤ Remaining = ₹6,000 ➤ Capital ratio = 1:1 → A & B get ₹3,000 each ➤ Final: A = ₹11,000, B = ₹3,000
3	**Commission to Working Partner**	Working Partner gets % of total profit as commission; rest divided in capital ratio	X (sleeping) invests ₹40,000, Y (working) invests ₹60,000. Y gets 10% commission. Profit = ₹50,000 ➤ Commission = ₹5,000 ➤ Remaining = ₹45,000 ➤ Capital ratio = 4:6 = 2:3 ➤ X = ₹18,000; Y = ₹27,000 + ₹5,000 = ₹32,000

Percentage

Basics:

Some commonly used percentage which can be easily converted into simple fractions.

$50\% = \frac{1}{2}$	$33\frac{1}{3}\% = \frac{1}{3}$	$66\frac{2}{3}\% = \frac{2}{3}$	$25\% = \frac{1}{4}$	$75\% = \frac{3}{4}$	$6\frac{1}{4}\% = \frac{1}{16}$
$20\% = \frac{1}{5}$	$40\% = \frac{2}{5}$	$60\% = \frac{3}{5}$	$80\% = \frac{4}{5}$	$16\frac{2}{3}\% = \frac{1}{6}$	$6\frac{2}{3}\% = \frac{1}{15}$
$33\frac{1}{3}\% = \frac{2}{6}$	$66\frac{2}{3}\% = \frac{4}{6}$	$83\frac{1}{3}\% = \frac{5}{6}$	$14\frac{2}{7}\% = \frac{1}{7}$	$28\frac{4}{7}\% = \frac{2}{7}$	$133\frac{1}{3}\% = \frac{4}{3}$
$12.5\% = \frac{1}{8}$	$37\frac{1}{2}\% = \frac{3}{8}$	$62.5\% = \frac{5}{8}$	$87.5\% = \frac{7}{8}$	$112.5\% = \frac{9}{8}$	$120\% = \frac{6}{5}$
$11\frac{1}{9}\% = \frac{1}{9}$	$22\frac{1}{9}\% = \frac{2}{9}$	$44\frac{4}{9}\% = \frac{4}{9}$	$55\frac{5}{9}\% = \frac{5}{9}$	$88\frac{8}{9}\% = \frac{8}{9}$	$125\% = \frac{5}{4}$
$9\frac{1}{11}\% = \frac{1}{11}$	$18\frac{2}{11}\% = \frac{2}{11}$	$36\frac{4}{11}\% = \frac{4}{11}$	$45\frac{5}{11}\% = \frac{5}{11}$	$90\frac{10}{11}\% = \frac{10}{11}$	$150\% = \frac{3}{2}$

Concept 1 Some Important Rules of Percentage:

- (a + b)% of x = a% of x + b% of x
- (a + b)% of (x + y) = a% of (x + y) + b% of (x + y)
- $Percentage\ increase = \frac{Increase\ in\ Quantity}{Original\ Quantity} \times 100$
- $Percentage\ decrease = \frac{Decrease\ in\ Quantity}{Original\ Quantity} \times 100$
- If any number is first increased by a% and then decreased by a% then net change will always be equal to $\frac{a^2}{100}$%

Concept 2 – Successive Percentage:

- If x% and y% change occurs successively, then overall change% $\left[x + y + \left(\frac{xy}{100}\right)\right]\%$
- If x%, y%, z% change occurs successively, then overall change% $\left[x + y + z + \frac{(xy + yz + zx)}{100} + \frac{(xyz)}{10000}\right]\%$

Concept 3- Price, Consumption & Expenditure

Basic Formula:

Price × Consumption = Expenditure

If expenditure is constant, then: **Price α 1 / Consumption**

This means — if price increases, consumption must decrease to keep expenditure the same, and vice versa.

Concepts based on Income, Expenditure and Savings:

Income = Expenditure + Savings

If income of 'B' is 'x%' more than that of 'A', then income of 'B' = Income of $A \times \{1 + (x/100)\}$

If income of 'B' is 'x%' less than that of 'A', then income of 'B' = Income of $A \times \{1 - (x/100)\}$

Concepts based on Population:

- When the present population of a town is 'X' and it increased/decreased at the rate of 'R'% each year, then the population of town after 'n' years -

  Population after n years = $X \times \left(1 \pm \frac{R}{100}\right)^n$
- When the present population of a town is 'X' and it increased/decreased at the rate of 'R'% each year, then the population of town 'n' years ago:

  Population n years ago = $\frac{X}{\left(1 \pm \frac{R}{100}\right)^n}$
- When the present population of a town is 'A' and it increased/decreased at different rates of x%, y% and z% successively in next three years, then the population of town after three years.

  Population of town= $A \times \left(1 \pm \frac{x}{100}\right) \times \left(1 \pm \frac{y}{100}\right) \times \left(1 \pm \frac{z}{100}\right)$
- When the present population of a town is 'A' and it increased by x%, decreased by y% and increased by z% in next three years respectively, then the population of town after three years.

  Population of town= $A \times \left(1 + \frac{x}{100}\right) \times \left(1 - \frac{y}{100}\right) \times \left(1 + \frac{z}{100}\right)$

Concepts based on Election:

◆ 1. **Votes Cast and Not Cast**

- Concept: A percentage of total voters cast their votes. The rest did not vote.
- Formula: $\boldsymbol{Votes\ cast = \frac{Percentage\ of\ votes\ cast}{100} \times Total\ voters}$

  Votes not cast =Total voters−Votes cast

◆ **2. Valid and Invalid Votes**

- Concept: Out of the votes cast, a certain percentage may be invalid.
- Formula: $\boldsymbol{Valid\ Votes = (\frac{Valid\ \%}{100}) \times Votes\ Cast}$

  Invalid votes=Votes cast−Valid votes

◆ **3. Winner Margin Based on Valid Votes**

- Concept: Two candidates receive a percentage of valid votes; the winner gets more.
- Formula: $\boldsymbol{Vote\ Margin = (\frac{Difference\ in\ \%\ of\ Valid\ Votes}{100}) \times Valid\ Votes}$

◆ **4. Finding Total Voters from Given Percentages and Margin**

- Reverse Calculation Concept:
  - Use the vote margin and percentage difference to work back and find the total votes or total voters.
- Formula: $\boldsymbol{Valid\ votes = \frac{Vote\ margin \times 100}{Percentage\ difference}}$

◆ **5. Percentage-based Comparison of Candidates**

- You may be asked:
  - What % more votes did A get than B?
  - What is the ratio of A's votes to B's?
- Formulas: $\boldsymbol{\%More = \frac{A's\ votes - B's\ votes}{B's\ votes} \times 100}$

$$\boldsymbol{Ratio = \frac{A's\ votes}{B's\ votes}}$$

Concepts based on Student Passing and Failure:

Sl. No.	Concept	Explanation / Formula	Example / Use
1	**Passing Marks**	Passing Marks = (P/100) × M *(P = required %, M = max marks)*	If passing % is 40 and max marks is 500 → Passing Marks = (40/100) × 500 = 200
2	**Failing by x marks**	Marks Obtained = Passing Marks – x	Student fails by 15 marks, passing marks = 200 → Obtained = 185
3	**Passing by y marks**	Marks Obtained = Passing Marks + y	Student passes by 20 marks, passing marks = 200 → Obtained = 220

Sl. No.	Concept	Explanation / Formula	Example / Use
5	**Finding Maximum Marks (M)**	Use: (P/100) × M ± margin = Given marks	A gets 25% and fails by 30 marks → (25/100)×M + 30 = Passing Marks
6	**Finding Passing %**	Use: Passing Marks = A ± margin = marks of student Use equations to find %	If A fails by 10 and B passes by 20, difference = 30, helps find A
7	**Conversion between Marks and %**	% = (Obtained / Max Marks) × 100 Marks = (% × Max) / 100	A scores 270 out of 600 → % = (270/600)×100 = 45%
8	**Reverse % Problems**	Given % and margin, use reverse % to find M	A fails by 10 marks at 40% → 40% of M = 260 ⇒ M = 650

PERCENTAGE & Ratio Proportion

Concepts based on Commission, Income Tax:

Commission It is a fee paid to an agent or salesperson for facilitating a sale.

Formula (Simple) **Commission = (Rate × Total Sales) / 100**

Commission on Profit If commission is based on profit: **Commission = (Rate × Profit) / 100**

Total Earnings **Total Income = Salary + Commission**

Types of Commission
1. **Fixed Commission** – Constant regardless of sales
2. **Variable Commission** – Changes based on sales slab or profit

Taxable Income Taxable Income = Total Income – Exemptions – Deductions

Tax Paid Tax Paid = Tax Rate × (Taxable Income in that slab)

Net Income (After Tax) Net Income = Gross Income – Income Tax

Concepts based on Ratio Proportion (Age & Number Based):

Concept	Key Point to Remember
1. Present Age Ratio	Let ages be: **a×k and b×k** (based on given ratio)
2. Age Difference	The difference in ages **always remains constant** over time
3. Future or Past Age Problems	Add/subtract same number of years to both ratios before comparing
4. Cross Multiplication	Use cross multiplication to solve equations involving ratio after 'x' years
5. Sum of Ages	Total Age = Sum of parts = (a + b + c...) × common multiple (k)
6. Age = Present ± Years	Age after n years = Present age + n Age n years ago = Present age – n
7. Watch for Tricky Language	Words like *'after', 'before', 'twice the age', 'thrice as old'* change framing
8. Frame Equations Carefully	Use "age + years" or "age – years" while maintaining the ratio correctly
9. Use one variable (x or k)	Convert ratios to **multiples of a variable** for easy substitution
10. Read units of age carefully	Check if the age is given in **years, months or a combination** (rare cases)

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Profit & Loss

Profit & Loss (Successive selling and Dishonest Dealings)

Concept	Formula / Shortcut	Explanation & Tips
Successive Profit or Loss	Net% = A + B + (A×B)/100	A and B are successive profit/loss percentages
Successive Loss Formula	Same: Net% = A + B + (A×B)/100 Use negative signs for loss	Example: 20% profit then 10% loss → Net% = 20 – 10 – (20×10)/100 = 8% profit
Dishonest Dealer (Using false weight)	Gain% = (Error in weight / True weight) × 100	If dealer gives 950g instead of 1kg: Gain = (50/950)×100
Dishonest Dealer (Mixing cheaper item)	Gain% = (Profit on mixing / Cost Price of mixture) × 100	E.g. mixing 1kg of ₹100 item with 1kg of ₹50 item & selling at ₹100
Double Selling (Profit in both deals)	If SP1 = SP2 but CP1 ≠ CP2: Loss% = ($Loss^2$ / Profit%)	Use when an item is sold at two different CPs but same SP
Overall Profit/Loss on Successive Selling	1st SP = CP + profit/loss, 2nd SP = new CP ± profit/loss	Always take base as **first CP**, then find overall %

Ratio and Proportion

Concepts based on Ratio Proportion (Age & Number Based):

Concept	Key Point to Remember
1. Present Age Ratio	Let ages be: **a×k and b×k** (based on given ratio)
2. Age Difference	The difference in ages **always remains constant** over time
3. Future or Past Age Problems	Add/subtract same number of years to both ratios before comparing
4. Cross Multiplication	Use cross multiplication to solve equations involving ratio after 'x' years
5. Sum of Ages	Total Age = Sum of parts = (a + b + c...) × common multiple (k)
6. Age = Present ± Years	Age after n years = Present age + n Age n years ago = Present age – n
7. Watch for Tricky Language	Words like *'after', 'before', 'twice the age', 'thrice as old'* change framing
8. Frame Equations Carefully	Use "age + years" or "age – years" while maintaining the ratio correctly
9. Use one variable (x or k)	Convert ratios to **multiples of a variable** for easy substitution
10. Read units of age carefully	Check if the age is given in **years, months or a combination** (rare cases)

Ratio Proportion

Concepts based on Coin:

S.No.	Concept	Explanation	Example
1	**Basic Ratio of Number of Coins**	Ratio given is based on **number of coins**, not value.	₹1:₹2:₹5 coins = 3:2:1 → Values = 3×1 + 2×2 + 1×5 = ₹12
2	**Ratio Based on Value**	Ratio given is based on **value**, must convert to number of coins using: Value ÷ Denomination	Value ratio 1:4:5 → ₹1x, ₹4x, ₹5x → No. of coins = x, 2x, x ⇒ Ratio = 1:2:1
3	**Total Number of Coins Given**	Use total parts of ratio to find coins of each denomination: Coins = (Ratio ÷ Total Ratio) × Total Coins	₹1:₹2:₹5 = 2:3:5, Total = 100 → Total parts = 10 → ₹1 coins = 20, ₹2 = 30, ₹5 = 50
4	**Total Value of Coins Given**	Use weighted sum formula: Total Value = (Qty × Denomination) for each coin	Ratio = 3:2:1, Total value = ₹360 → 3x + 4x + 5x = 12x = 360 ⇒ x = 30 → Coins = 90, 60, 30
5	**Difference-Based Questions**	Use difference in value or count to form equations and solve for ratio parts.	Ratio of ₹1:₹2 = 3:4, Value difference = ₹10 → 8x – 3x = 5x = 10 ⇒ x = 2 ⇒ ₹1 coins = 6, ₹2 = 8
6	**Mixture or Replacement Type**	When coin counts are added/removed, resulting in a new ratio; use ratio-change equation.	₹1:₹2 = 2:5; add 10 ₹1 coins → New ratio = 4:5 ⇒ (2x + 10)/5x = 4/5 ⇒ solve for x

Simple and Compound Interest

S.No.	Concept / Type	Formula / Explanation	Usage / Exam Tip
1	**CI vs SI for 2 years**	Difference = $P \times (R/100)^2$	Useful shortcut when time = 2 years
2	**CI vs SI for 3 years (shortcut)**	Difference = $P \times R^2(300 + R)/100^3$	Direct formula for quick solving
3	**When CI = SI**	Only happens when R = 0% or T = 1 year	Conceptual question often asked
4	**Time when SI and CI difference known**	T = 2 → Use P = Diff × 100^2 / R^2	Reverse use of 2-year shortcut formula
5	**Finding rate (R) from SI and CI difference**	R = √(100 × Diff / P)	Valid only for 2 years

Simplification

Key Concepts and Conditions:

BODMAS Rule:

Order of operations: Brackets, Orders (Powers and Roots), Division, Multiplication, Addition, and Subtraction:

Preference Order	Operator	Symbol
1st	Brackets	( ) , { } , [ ]
2nd	Orders (Powers and Roots)	^, √
3rd	Division	÷
4th	Multiplication	×
5th	Addition	+
6th	Subtraction	-

Understand the rules of sign operations:

(+) × (+) = (+)

(-) × (-) = (+)

(-) × (+) = (-)

Important Formulas:

Fractions:

Addition/Subtraction: $\frac{a}{b} + \frac{c}{d} = \frac{ad+cb}{bd}$

Multiplication: $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$

Division: $\frac{a}{b} \div \frac{c}{d} = \frac{a \times c}{b \times d}$

Speed Time and Distance

Average Speed

Topic	Formula / Concept	Explanation
Basic Formula	Speed = Distance / Time	Use this to find any one of speed, distance, or time if the other two are known.
Units Conversion	1 km = 1000 m 1 hour = 60 minutes = 3600 seconds	Useful for converting units as needed.
	1 km/hr = (5/18) m/sec 1 m/sec = (18/5) km/hr	To convert speed units.
Average Speed (general)	Average Speed = Total Distance / Total Time	Used when total distance and total time are known.
Two Equal Distances	Average Speed = $(2 \times S_1 \times S_2) / (S_1 + S_2)$	Used when the distance is the same but speed is different for each leg.
Different Distances	Average Speed = $(D_1 + D_2) / ((D_1/S_1) + (D_2/S_2))$	General formula for two parts with different distances and speeds.
If time difference is given	Distance = $(S_1 \times S_2 / (S_2 - S_1)) \times$ Time difference	Useful when two persons travel same distance at different speeds, and time difference is given.

Relative Speed

Concept	Formula / Explanation	Example
1. Relative Speed (Same Direction)	Relative Speed = $Speed_1 - Speed_2$	Two trains at 60 km/hr and 40 km/hr in the same direction ⇒ Relative Speed = **20 km/hr**
2. Relative Speed (Opposite Direction)	Relative Speed = $Speed_1 + Speed_2$	Two trains at 60 km/hr and 40 km/hr in opposite directions ⇒ Relative Speed = **100 km/hr**
3. Time to Cross Each Other (Two Bodies Moving)	Time = Distance between bodies / Relative Speed	Two trains of length 120 m and 80 m moving opposite at 30 m/s and 10 m/s:Total distance = 200 mRelative speed = 40 m/s ⇒ Time = 200/40 = **5 sec**
4. Overtaking / Meeting Time (Relative Speed)	Use relative speed depending on direction	Two bikes, 60 km/hr and 40 km/hr, same direction: Distance = 100 km ⇒ Time = 100 / (60−40) = **5 hr**
5. Average Speed (Same Distance)	Avg Speed = 2ab / (a + b), where a and b are two different speeds for equal distance	Speed to go = 60 km/hr, return = 40 km/hr:Avg speed = 2×60×40 / (60+40) = 4800/100 = **48 km/hr**

Problems on Train

Concept	Formula	Explanation	Example
1. Train crossing a pole	Time = Length of train / Speed	Pole is point-sized – only train's length matters.	180 m train at 60 km/h → 60×5/18 = 16.67 m/s → Time = 180/16.67 ≈ **10.8 sec**
2. Train crossing a platform	Time = (Train length + Platform length) / Speed	Train must cover own length + platform.	Train = 120m, Platform = 180m, Speed = 54 km/h → Speed = 15 m/s → Time = (120+180)/15 = **20 sec**
3. Two trains crossing each other (same direction)	Time = $(L_1 + L_2) / (S_1 - S_2)$	Relative speed when trains move **in same direction**.	Train A = 150m, 60 km/h; Train B = 100m, 40 km/h → Speed diff = 20 km/h = 5.56 m/s → Time = (150+100)/5.56 ≈ **45 sec**
4. Two trains crossing each other (opposite direction)	Time = $(L_1 + L_2) / (S_1 + S_2)$	Relative speed when trains move **in opposite direction**.	Train A = 180m, 72 km/h; Train B = 120m, 54 km/h → Total speed = 126 km/h = 35 m/s → Time = (180+120)/35 = **8.57 sec**

Concept	Formula	Explanation	Example
5. Train crossing a man (same/opposite direction)	Time = Length of train / Relative speed	Man has speed. Use relative speed concept.	Train = 180m, 72 km/h; Man = 6 km/h, same dir → Speed = 66 km/h = 18.33 m/s → Time = 180/18.33 ≈ **9.8 sec**
6. Length of train	Length = Speed × Time (in m/s × sec)	Direct multiplication if time and speed known.	Speed = 20 m/s, Time = 12 sec → Length = 20×12 = **240 m**
7. Average Speed	Avg. Speed = $(2 \times S_1 \times S_2) / (S_1 + S_2)$	Only for same distance with two speeds.	60 km/h & 90 km/h → Avg. = (2×60×90)/(60+90) = **72 km/h**

Race

Concept	Formula / Description	Example
Head start (Distance)	If A beats B by d meters, it means B runs D – d meters when A runs D.	A covers 100 m, B covers 90 m ⇒ A beats B by 10 m
Time-based lead	If A beats B by t seconds, calculate who is faster based on their time to finish total race	Compare total times to complete race
Ratio of speeds	If A beats B by d meters in a D-meter race: Speed_A / Speed_B = D / (D – d)	Use when distance difference is known
Equal Time, Different Distances	Ratio of speeds = Ratio of distances	A covers 100 m, B covers 80 m in same time ⇒ Speed A : Speed B = 5:4
Equal Distance, Different Time	Ratio of speeds = Inverse of time ratio	If A takes 10 min, B takes 12 min ⇒ Speed A : B = 6:5
Circular track (Meeting Point)	If two people run around a circle of length L: Meeting time = L / Relative Speed	Adjust relative speed based on direction
Catch-up time (A chasing B)	Time = Distance gap / Relative Speed	For overtaking or closing the distance

Time and Work

Time and Work: Work and Efficiency Concepts

Concept	Formula / Explanation	Example / Note
Work	Work = Time × Efficiency	If A does 5 units/day for 6 days, Work = 5 × 6 = 30 units
Efficiency	Efficiency = Work / Time	If work is 60 units and A takes 10 days, Efficiency = 6 units/day
Time	Time = Work / Efficiency	If work = 48 units, efficiency = 6 units/day, Time = 8 days
Total Work	Total Work = LCM of individual times	If A takes 6 days, B takes 8 days, Total Work = LCM(6,8) = 24 units
Combined Efficiency	A's Eff + B's Eff = Total Eff	If A = 4 units/day, B = 6 units/day, together = 10 units/day
Efficiency Ratio (based on time)	Efficiency $\propto 1/\text{Time} \Rightarrow E_1 : E_2 = T_2 : T_1$	If A takes 5 days, B takes 10 days $\Rightarrow E_1 : E_2 =$ 10:5 = 2:1
Work Ratio (based on efficiency)	Work done $\propto$ Efficiency × Time	Use to split wages/amounts based on contribution
Men, Days, Work	$M_1 \times D_1 = M_2 \times D_2$ (when work is same)	Used when number of men and days change
One day's work	1 day's work = 1 / total days	If A finishes in 10 days, 1 day work = 1/10
Total work from 1 day's work	Total Work = 1 day's work × Total Days	Reverse of above
A is 'x%' more efficient than B	Let B's eff = 100, then A = 100+x $\Rightarrow$ Ratio = A:B = (100+x):100	If A is 25% more efficient $\Rightarrow$ Ratio = 125:100 = 5:4
A is 'x%' less efficient than B	A = 100−x $\Rightarrow$ Ratio = A:B = (100−x):100	If A is 20% less $\Rightarrow$ Ratio = 80:100 = 4:5
Alternate Work Days	Use unit method or LCM, consider number of days each person works	Alternate day working is solved step-wise
Pipe & Cistern as Negative Work	Leakage = negative efficiency	If pipe fills 10 units/day and leak empties 2, net = 8 units/day
Work Sharing Based on Efficiency	Share $\propto$ Efficiency × Time worked	For unequal working durations
Inversely Proportional Time	Time $\propto$ 1 / Efficiency	Time taken is inversely proportional to speed or efficiency

Time and Work – Concepts, Formulas & Shortcuts

Concept / Topic	Details / Explanation	Formula / Key Point
1. Basic Work Formula	Work is directly proportional to time and efficiency.	Work = Time × Efficiency
2. Unit Work Method	Total work is considered as 1 unit (or LCM of days if needed).	Use unitary method: If A completes in 10 days, A's 1 day work = 1/10

Concept / Topic	Details / Explanation	Formula / Key Point
3. Time Taken to Complete Work	If A can do a piece of work in 'x' days.	Work done/day = 1/x ⇒ Total Time = 1 ÷ Work done/day
4. Combined Work	If A can do a work in x days and B in y days.	(A + B)'s 1 day work = 1/x + 1/y Time = xy / (x + y)
5. Efficiency and Time Relation	More efficiency means less time.	Efficiency ∝ 1 / Time
6. Ratio of Work and Time	Compare work done by different people.	$Time_1$ / $Time_2$ = $Efficiency_2$ / $Efficiency_1$
7. Work Done in 'n' Days	If A's 1 day work is 1/x, work done in n days = n/x	Total Work Done = Days × Efficiency
8. Remaining Work	Total work - work done = Remaining work	Can use unit work method or LCM
9. A leaves the work after n days	Handle work in parts: work done with A + rest by others	Break the problem into phases
10. Men, Women, and Children	Use efficiency ratio (e.g., M:W:C = 4:3:2)	Convert all to a single unit for comparison
11. LCM Method	Take LCM of individual days to assume total work	Makes calculation easier
13. Work and Wages	Wages ∝ Work Done (not time)	Total Wages × (Person's Work / Total Work)
14. Unequal Working Days (Wages)	If people work different days, calculate work individually	Assign wages in ratio of work done
15. Alternate Working Days	A & B work alternately (e.g., A-B-A...)	Count days and multiply by individual 1 day work
16. Negative Work (Leaking Pipes / Spoilers)	If B undoes A's work, subtract	Net work = A's work – B's work
17. Work Completion %	To find part of work completed or left	Work done = (No. of days × 1 day work)

Work and Wages Specific Concepts

Case	Explanation	Formula / Example
Equal Wages	All do equal work	Divide equally
Unequal Efficiency	More efficient person gets more wage	Wages ∝ Work Done
Different Time Periods	Different days of work	Wages ∝ (Days × Efficiency)
Ratio-Based Wages	If A:B:C work ratio = 2:3:5	Divide total wages in 2:3:5
Wages with Partial Work	If total work = 60 units A: 20 units, B: 40 units	A's Wage = (20/60) × Total Wage

Time and Work – (Man-Day Based)

S.No.	Concept	Explanation / Formula	Example / Note
1.	**Work and Time Relationship**	More men → less time; More time → more work	Work ∝ Men × Days
2.	**Basic Formula**	Work = Men × Days × Hours (if hourly rate given)	If 5 men do a work in 4 days, total work = 5×4 = 20 man-days
3.	**Combined Work Rate**	A's 1-day work + B's 1-day work = (A + B)'s 1-day work	If A = 1/10, B = 1/15, together = 1/10 + 1/15 = 1/6
4.	**Total Work in Units**	If A completes in x days, then total work = x units	Used to avoid fractions
5.	**Work Efficiency Ratio**	If A : B = efficiency ratio, then Time ratio = 1 : efficiency	If A is twice as efficient as B → Time ratio = 1 : 2
6.	**More Men, Less Time**	$M_1 \times D_1 = M_2 \times D_2$ (if work constant)	Men × Days = Constant
7.	**A and B together, but A leaves after some days**	Use A + B's rate × days + B's rate × remaining days	Useful for replacement problems
8.	**Inversely Proportional Rule**	Work is constant → Men and Days are inversely proportional	$M_1 : M_2 = D_2 : D_1$
9.	**Negative Work (Leakage, Destruction)**	Net work = Work done – Work undone	Pipes filling and leakage

S.No.	Concept	Explanation / Formula	Example / Note
10.	**LCM Method**	Assume total work as LCM of days taken by individuals	Simplifies fraction-based calculations
11.	**One finishes a fraction, rest completes**	Find part done, remaining part and respective time	Very common in SSC exams

Time and Work – (Alternate Days Working Concept)

Type of Question	Example Scenario	How to Approach
1. A starts, B next, alternate till completion	A can do a work in 6 days, B in 12 days. They work alternately starting with A. In how many days will the work be completed?	Find A's and B's 1-day work. Add both for 2-day cycle. Repeat till total work is done. If work left after full cycles, assign last day accordingly.
2. Total work given, who works on last day?	Work is 20 units. A does 4 units/day, B does 2. A starts alternately. Who works on last day?	Find how much work is done in each 2-day cycle (A+B), divide total work, check remaining. Remaining work tells who works last.
3. B starts instead of A	B starts first, works alternate days with A. Who finishes last?	Same logic as above, but reverse the sequence. Start cycle with B.
4. Work completed in N days, how many days did A and B work?	A and B work on alternate days starting with A. Total work done in 7 days. How many days did each work?	Count A's and B's days in sequence: A works on odd days, B on even. Split 7 days accordingly: A = 4 days, B = 3 days.
5. Unequal Efficiency / Units Given	A does 5 units/day, B does 3 units/day. A starts, they work alternately. Total work = 35 units. Find days needed.	2-day work = A+B = 8 units. Divide 35 by 8. After full cycles, assign leftover work to A or B as per day.

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